

TrES-3b

R-2771

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160711 · created 2026-09-03T23:09:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457580.691 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.015
Transit depth (Rp/Rs)^2	3.02 +/- 0.54 [%]
Orbital Inclination (inc)	82.95 +/- 0.96
Airmass coefficient 1 (a1)	1.167 +/- 0.012
Airmass coefficient 2 (a2)	-0.156 +/- 0.084
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	None
Optimal Aperture	4.06
Optimal Annulus	17.94
Transit Duration (day)	0.0586 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.015 vs 0.16309 (deviation 0.73σ)
Duration fit vs published	0.0586 d vs 0.0567 d (deviation 0.44σ)
Mid-time O–C	1.7 min
Depth SNR	11.6
Residual scatter	1.01 %

Light curve

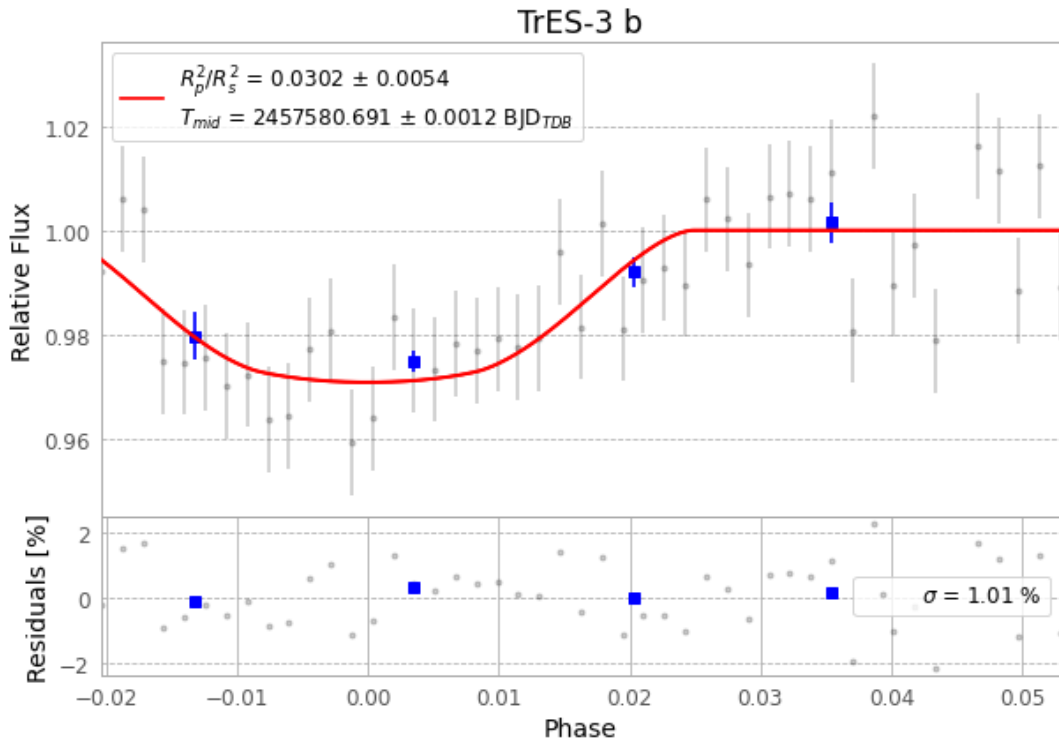


Figure 1 — FinalLightCurve_TrES-3 b_2016-07-10.png (SHA-256 9a87de12b49cbaf6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [361, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 443702466095f2ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

47 FITS frames (31 MB), TRES-3160711035112.FITS → TRES-3160711060912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160711.log (7d79e12a7e51...), FinalLightCurve_TrES-3 b_2016-07-10.png (9a87de12b49c...), FinalLightCurve_TrES-3 b_2016-07-10.csv (7883d46f404d...)

Compute resources

Wall time 237.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:09Z.